returning the top element without removing it from the container.

D.3 Queue

A queue is a collection that is based on the FIFO (First In First Out) policy. Therefore, elements of a queue will be returned in the same order as they were added to it. While queues are also limited access data structures, elements are added to the front of the queue and removed from its back (which also means their implementation is slightly more complicated than stacks').

A queue is, as you can imagine at this point, a particular case of priority queue where the priority of an element is its age: older elements have the highest priority.

The name, in this case, is quite revealing, and indeed in the real world this is the DS we (subconsciously) use every time we are waiting in a line: whoever has been waiting the longest is served first. Like stacks, queues are fundamental in computer science, and indeed they are used in many algorithms.